

YEHUB

Youth Enterprise Hubs

Technical Documentation & Site Guide

A Citrus_alu Initiative

African Leadership University · RWBISD E-LAB 2026

Version 1.0 · July 2026

Empowering South Sudan's Youth to Build Businesses and Create Jobs.

1. Overview

What is YEHUB?

YEHUB (Youth Enterprise Hubs) is a web-based platform prototype designed as part of the Citrus_alu team's RWBISD E-LAB challenge at African Leadership University. It simulates a complete digital ecosystem for community-based entrepreneurship centres serving unemployed youth aged 18–35 in South Sudan.

The site is built in React and powered by Citrus AI — a Claude-based career coaching assistant customised specifically for South Sudan's youth entrepreneurship context. It serves as the prototype requirement for the E-LAB Digital Print rubric and demonstrates how the Youth Enterprise Hubs concept would work in practice.

Core Mission

To equip young South Sudanese people aged 18–35 with practical skills, mentorship, funding access, and market connections — transforming job seekers into job creators through an integrated, community-driven hub model.

Key Statistics

Target Users	Young people aged 18–35 in South Sudan
Primary Goal	Reduce youth unemployment through entrepreneurship support
Grant Target	\$10,000 seed fund to establish the first YEHub
Hub Stages	7 stages: Register → Assess → Train → Incubate → Fund → Launch → Market
AI Feature	Citrus AI — personalised career coach powered by Claude (Anthropic)
Platform	Web application (React + Vite), deployed on Vercel

2. Technology Stack

Framework	React 18 — component-based UI library
Build Tool	Vite 5 — fast development server and production bundler
Styling	CSS-in-JS (inline styles) — no external CSS library required
Fonts	Inter (all weights) via Google Fonts CDN
AI Integration	Anthropic Claude API (claude-sonnet-4-6) — Citrus AI chat
Deployment	Vercel — automatic CI/CD from GitHub
State Management	React useState / useEffect — no Redux or external state library
Routing	Page-based state switching (no React Router needed)
Auth	Simulated — React state only (no backend). Demo mode included.

Project File Structure

yehub/
■■■ public/
■ ■■■ favicon.svg
■■■ src/
■ ■■■ App.jsx ← Full application (975 lines)
■ ■■■ main.jsx ← React entry point
■ ■■■ index.css ← Global resets + keyframe animations
■■■ index.html ← HTML shell with Google Fonts
■■■ package.json ← Dependencies (React, Vite)
■■■ vite.config.js ← Build configuration

3. Pages & Navigation

YEHUB has two navigation contexts: Public (unauthenticated) and Authenticated (logged in). All navigation is handled by React state — no page reloads occur.

Public Navigation (8 pages)

Home	Hero with auto-advancing carousel, stats, journey preview, skills grid, Citrus AI teaser, CTA sections.
About	YEHUB mission, vision, approach, primary/secondary beneficiaries, and revenue sustainability model.
Programs	6 hub programmes: Training, Incubation, Workspace, Finance, Market Linkages, Community Learning. Each has expandable details.
Skills	4 skill categories with all individual courses. Tab-based filtering. Each course shows level, duration, and enrol button.
Stories	3 success story cards (Amara, David, Grace) with before/after income data and direct quotes.
Partners	6 partner types with descriptions. Includes a 'Become a Partner' CTA section.
Community	4 community features + live-style forum posts with tags, likes, and replies.
Contact	Contact form (with sent confirmation state) + location, email, phone, and hours info.

Authenticated Navigation (3 pages)

Dashboard	Welcome header with user name and current stage. Progress tracker, 4 stat cards, task list, Citrus AI shortcut.
My Journey	Interactive 7-stage progression tracker. Each stage has checkable activities, downloadable resources, and Next/Previous navigation.
Citrus AI	Full-page Claude-powered chat interface with quick-reply suggestions, typing indicator, and clear conversation button.

4. Authentication System

Authentication is fully simulated using React state. No backend, database, or server is required. This is by design — the site is a prototype demonstrating the YEHUB concept, not a production application.

Auth Modal

Clicking 'Log In' or 'Join Free' opens a modal overlay (not a separate page). The modal contains three modes:

- **Log In** — Email + password form. Any input is accepted in demo mode.
- **Join Free** — 3-step registration: personal info → interests → goal. Each step validates before advancing.
- **Try Demo** — One-click login as 'Demo Explorer' with stage 2 pre-loaded. No form required.

User State Shape

```
const user = {  
  name: "string – full name",  
  email: "string – email address",  
  age: "string – age (optional)",  
  location: "string – city/country (optional)",  
  interest: "string – skill category chosen",  
  goal: "string – primary entrepreneurship goal",  
  stage: "number – current hub stage (0-6)",  
  isDemo: "boolean – true for demo users"  
}
```

5. Interactive Stage Progression

The 'My Journey' page is the core interactive feature of the site. It guides members through 7 clearly defined stages, with each stage containing activities and resources.

Stage 0 — Register	1 day · Create profile, choose interests, sign guidelines, get member ID.
Stage 1 — Assess	Days 2–4 · Digital literacy test, business quiz, strengths survey, advisor session.
Stage 2 — Train	Weeks 1–6 · Daily training sessions, weekly assignments, peer discussions, mid-assessment.
Stage 3 — Incubate	Weeks 7–10 · Business model canvas, business plan, mentor pitch, market research.
Stage 4 — Fund	Weeks 11–12 · Seed fund application, microfinance meetings, grants exploration, financial projections.
Stage 5 — Launch	Week 13 · Business registration, marketing campaign, marketplace store setup, launch event.
Stage 6 — Market	Ongoing · Customer expansion, monthly AI health check, alumni mentorship, innovation competitions.

Each stage page includes: (a) checkable activity list with progress bar, (b) resource list with download buttons, (c) a Citrus AI tip panel, and (d) Previous / Next stage navigation buttons.

6. Citrus AI — The Career Coach

Citrus AI is YEHUB's most distinctive feature — a real AI-powered career coaching assistant built on Anthropic's Claude API. It is not a generic chatbot. It is specifically configured for South Sudan's youth entrepreneurship context.

How It Works

1. The user types a message and presses Enter or the send button.
2. The message is sent to the Anthropic Claude API (claude-sonnet-4-6).
3. A custom system prompt configures Claude to act as Citrus AI — focused on South Sudan markets, YEHUB stages, and entrepreneurship coaching.
4. The user's profile (name, interest, stage) is appended to the system prompt for personalisation.
5. Claude's response is streamed back and displayed in the chat interface.
6. Quick-reply buttons suggest common questions to guide the conversation.

System Prompt Summary

The Citrus AI system prompt instructs Claude to:

- Act as a warm, encouraging, practical entrepreneurship coach
- Create personalised business roadmaps based on user interests
- Recommend skills relevant to South Sudan's local market (agriculture, solar, tailoring, phone repair, digital services)
- Guide users through the 7 YEHUB stages with specific action steps
- Generate monthly Business Health Scores (out of 100) with recommendations when asked
- Keep responses concise — under 200 words — with bullet points and emojis
- Always end with one clear next action step

API Call Structure

```
fetch("https://api.anthropic.com/v1/messages", {  
  method: "POST",  
  headers: { "Content-Type": "application/json" },  
  body: JSON.stringify({  
    model: "claude-sonnet-4-6",  
    max_tokens: 1000,
```

```
system: SYSTEM_PROMPT + userContext,
```

```
messages: conversationHistory
```

```
})
```

```
})
```


7. Deployment Guide

Step 1 — Push to GitHub

```
cd yehub
git init
git add .
git commit -m "Initial commit – YEHUB prototype"
git remote add origin https://github.com/YOUR_USERNAME/citrus-project.git
git push -u origin main
```

Step 2 — Deploy on Vercel

1. Go to vercel.com and log in with your GitHub account.
2. Click 'New Project' and import your GitHub repository.
3. Set the Root Directory to **yehub** (not the root of the monorepo).
4. Leave all other settings as default — Vercel detects Vite automatically.
5. Click Deploy. Your site will be live at a `.vercel.app` URL within 60 seconds.

Step 3 — Update Cross-Site Links

After both sites are deployed, update these two lines so they link to each other:

```
// In yehub/src/App.jsx (line 8):
const CITRUS_URL = "https://YOUR-CITRUS-ALU-URL.vercel.app";

// In citrus-alu/src/App.jsx (line 9):
const YEHUB_URL = "https://YOUR-YEHUB-URL.vercel.app";
```

Environment Note

The Citrus AI (Claude API) calls are made directly from the browser. The Anthropic API key is handled automatically by the Claude.ai artifact environment during development. For production deployment, you will need to add your own `ANTHROPIC_API_KEY` via a backend proxy or Vercel Edge Function to keep the key secure.

8. Running Locally

Prerequisites

Node.js	Version 18 or higher — download from nodejs.org
npm	Comes with Node.js — used to install dependencies
Git	For cloning and version control
Browser	Chrome, Firefox, or Edge (modern version)

Installation & Start

```
# Clone the repository
git clone https://github.com/YOUR_USERNAME/citrus-project.git
cd citrus-project

# Install and run YEHUB
cd yehub
npm install
npm run dev
# → Opens at http://localhost:5173

# In a new terminal – Install and run Citrus_alu
cd ../citrus-alu
npm install
npm run dev
# → Opens at http://localhost:5174
```

9. Complete Feature Summary

Hero Carousel	Auto-advancing image slider with fade transitions, dot indicators, and arrow navigation.
Auth Modal	Popup login/register with 3-step registration flow and one-click demo mode.
Demo Mode	Explore the full site without registering — labelled clearly in the navbar.
Dashboard	Personalised welcome, progress tracker, stat cards, task list, and AI shortcut.
Stage Progression	Interactive 7-stage journey with checkable activities, progress bars, and Next/Prev navigation.
Citrus AI Chat	Full Claude-powered chat with quick replies, typing indicator, and clear button.
Skills Browser	4 categories, 19 courses with level badges, duration, descriptions, and enrol buttons.
Programs Page	6 hub programmes with expandable details and Apply Now buttons.
Success Stories	3 entrepreneur stories with before/after data and income figures.
Partners Page	6 partner types + Become a Partner CTA.
Community Forum	Forum posts with tags, likes, replies preview.
Contact Form	Live form with sent confirmation state.
Responsive Design	Mobile-first layout using CSS Grid and Flexbox with auto-fit columns.
Reduced Motion	All animations respect prefers-reduced-motion media query.
Page Transitions	Smooth opacity fade between all page navigations.

10. Project Team

Achol	Team Leader — strategic vision and final decisions
Elysee	Operations Manager — timelines, coordination, and delivery
Ian	Secretary — documentation and team communications

Frank	Editor — content, design direction, and website development
Eden	Researcher — evidence gathering, interviews, and field research

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